

EduPilot AI

Adaptive Learning Platform with AI-Powered Learning Path Optimization

Comprehensive Project Documentation

1. Project Overview

EduPilot AI is an AI-powered adaptive learning platform designed to improve the teaching-learning process by providing personalized learning experiences. Unlike conventional AI chatbots, the platform understands a student's current knowledge level, identifies conceptual gaps, and generates personalized learning paths using teacher-provided content.

2. Problem Statement

Current educational platforms provide the same learning material to every student regardless of their prior knowledge. General-purpose AI assistants answer questions but do not understand prerequisite concepts, monitor learning progress, or provide structured learning paths. Faculty also lack insights into class-wide misconceptions and learning analytics.

3. Objectives

- Personalize learning for every student.
- Detect knowledge gaps automatically.
- Generate prerequisite-aware learning paths.
- Answer questions only from teacher-uploaded material.
- Provide adaptive explanations based on student proficiency.
- Offer actionable analytics to teachers.

4. How the System Works

1. Teacher uploads notes, PDFs, PPTs and assignments.
2. Documents are parsed and indexed.
3. AI extracts concepts and builds a prerequisite knowledge graph.
4. Students take quizzes and ask questions.
5. The system estimates concept mastery.
6. The Learning Path Optimizer recommends the next topics to study.
7. The AI Tutor retrieves relevant course material and generates personalized responses.
8. Analytics dashboards help teachers monitor progress.

5. Main Modules

Teacher Dashboard: Upload content, manage courses, view analytics.

Student Dashboard: Chat with AI, quizzes, progress tracking.

Knowledge Graph Engine: Builds concept dependencies.

Learning Path Optimizer: Generates personalized study plans.

Adaptive AI Tutor: Provides context-aware explanations.

Analytics Module: Visualizes mastery, weak concepts, and engagement.

6. AI Components

- Retrieval-Augmented Generation (RAG)
- Vector Database (FAISS/ChromaDB)
- Embedding Model
- Local Open-Source LLM (Qwen/Gemma/Phi)
- Knowledge Graph (NetworkX)
- Concept Extraction Pipeline
- Student Knowledge Model
- Learning Path Recommendation Engine

7. Technology Stack

Frontend: Next.js, React, Tailwind CSS

Backend: FastAPI, Python

Database: PostgreSQL

AI Libraries: Hugging Face Transformers, Sentence Transformers, NetworkX, FAISS/ChromaDB, LangChain (optional)

Deployment: Docker, Vercel (Frontend), Render/AWS (Backend)

8. AI Syllabus Mapping

Intelligent Agents: Adaptive AI tutor.

Search: Learning path traversal and retrieval.

Knowledge Representation: Knowledge graph.

Planning: Personalized learning roadmap.

Information Retrieval: RAG pipeline.

Information Extraction: Concept extraction from documents.

NLP: Question answering and summarization.

9. Innovation

Unlike traditional educational chatbots, EduPilot AI combines Retrieval-Augmented Generation, Knowledge Graphs, Student Knowledge Modelling, Adaptive Explanations, Personalized Learning Paths, and Teacher Analytics into a single intelligent educational platform. The learning path is generated algorithmically using prerequisite relationships rather than simply relying on an LLM response.

10. Development Phases

Phase 1: Project setup, authentication, teacher/student dashboards.

Phase 2: Document upload and RAG pipeline.

Phase 3: Knowledge graph generation and visualization.

Phase 4: Learning Path Optimizer and adaptive tutor.

Phase 5: Analytics dashboard and testing.

Phase 6: Deployment, evaluation and presentation.

11. Expected Deliverables

Working web application, AI tutor, teacher dashboard, student dashboard, knowledge graph visualization, personalized learning path generator, quiz generation, analytics dashboard, project report, presentation, and source code.

12. Future Scope

Support multiple subjects, multimodal learning (images/audio), spaced repetition, coding tutor, institutional deployment, predictive performance analytics, and multilingual educational assistance.